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AI anxiety and adoption intention in higher education based on an extended TAM-UTAUT and PLS-SEM analysis

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This study aims to deconstruct the complex relationship between artificial intelligence anxiety and generative AI adoption intention in the context of higher education. An extended analytical framework is constructed by integrating the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT). Taking faculty and students from three universities in Sichuan Province as the research objects, data are collected through a questionnaire survey, and empirical research is conducted using the Partial Least Squares Structural Equation Modeling (PLS-SEM). Focusing on three dimensions of AI anxiety—AI learning anxiety, AI sociotechnical blindness anxiety, and AI job displacement anxiety—the study systematically examines their indirect impact paths on adoption intention through mediating variables such as performance expectancy and effort expectancy, and investigates the moderating effects of disciplinary background and AI self-efficacy. The findings reveal that AI anxiety exhibits a significant "double-edged sword" effect on generative AI adoption intention: AI job displacement anxiety comprehensively inhibits the core variables of technology acceptance; in contrast, AI sociotechnical blindness anxiety not only positively promotes effort expectancy and adoption intention but also negatively affects performance expectancy. In the technology acceptance mechanism, effort expectancy and social influence are the factors driving adoption intention, while disciplinary background and AI self-efficacy regulate the transmission paths of anxiety by shaping cognitive paradigms and psychological resources. The theoretical value of this study lies in breaking through the simplistic inhibition hypothesis of emotional factors in technology acceptance research. Practically, it provides empirical evidence for higher education institutions to formulate differentiated AI training strategies.

Keywords AI anxiety, Adoption intention, Higher education, Extended TAM-UTAUT, PLS-SEM

The rapid advancement of Generative Artificial Intelligence (GAI) is profoundly reshaping the teaching ecology and knowledge production models in higher education[1]. From intelligent tutoring systems designing personalized learning pathways[2] to natural language processing technologies assisting academic writing and literature analysis[3], GAI has demonstrated enormous potential to enhance teaching efficiency[4] and expand cognitive boundaries[5]. According to Deloitte's 2024 Global EdTech Trends Report, 92% of universities worldwide have incorporated GAI into their strategic development plans[6], with the adoption rate of GAI tools in higher education institutions increasing by 215% year-on-year between 2023 and 2024. Among these, the fastest growth in applications has been observed in teaching assistance and research innovation. Meanwhile, an empirical study published by the Stanford University Education Technology Laboratory in 2024 indicated that the acceptance of GAI among the higher education community exhibits significant stratification: 41% of faculty and students regard it as a "cognitive enhancement tool" and actively explore application scenarios, while 37% express multiple concerns regarding technical reliability, academic ethical risks, and impacts on professional value[7]. This cognitive differentiation reflects the dynamic game of "anxiety-adaptation" in the diffusion of intelligent technologies, and also highlights the theoretical and practical urgency of systematically deconstructing the complex mechanism between AI anxiety and adoption intention[8].

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In the field of technology acceptance research, classic frameworks such as the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) have profoundly revealed the driving role of rational cognitive factors — including performance expectancy and social influence — in technology adoption[9,10]. As the latest extension of the UTAUT series, UTAUT3 has significantly improved the explanatory power for acceptance behaviors of emerging intelligent technologies by adding constructs of personal innovativeness, perceived trust, perceived risk, and habit, and has demonstrated broad adaptability in empirical studies across multiple fields.

From the perspective of existing research progress, the application of UTAUT3 presents three core directions. In the context of educational technology, Gupta et al. explored educators' adoption intention of virtual communication technology based on UTAUT3 and found that the interaction effect between personal personal innovativeness and social influence could explain the variance in adoption intention[11], while teachers' technology anxiety would weaken the driving role of effort expectancy; their subsequent study on adolescents' adoption intention of multiplayer mobile games[12] further confirmed that the effect intensity of the habit construct in sustained technology use was significantly higher than that of traditional performance expectancy; Salam et al. combined UTAUT3 with the MOOC utilization model and pointed out that in the digital scenario of higher education, perceived trust — especially trust in the accuracy of technical content — serves as a key mediator connecting performance expectancy and adoption behavior, but their study did not involve the interference of emotional factors on trust formation[13].

In the context of GAI and intelligent systems, Ishengoma and Shao applied UTAUT3 to the research on AIoT system integration in libraries and found that the negative effect of risk perception on adoption intention needed to be alleviated through the "trust-habit" chain mediation[14]; Khan et al.'s meta-analysis on mobile learning in higher education showed that although the average explanatory power of UTAUT3 in the field of educational AI was higher than that of UTAUT2, the integration of emotional variables remained in a fragmented stage[15], but they did not focus on knowledge-intensive scenarios such as higher education, making it difficult to directly transfer the findings to GAI. In the context of other service fields, Laradi et al. confirmed in their study on the continuous use of mobile banking that the interaction effect between the habit construct of UTAUT3 and attitude could effectively predict long-term adoption behavior[12]; Ngusie et al. emphasized in their study on electronic health record systems that the effect intensity of facilitating conditions in resource-constrained regions was significantly lower than that in developed regions[16], which provided a reference for understanding regional differences but did not involve the moderating role of group cognitive differences such as disciplinary background.

Beyond UTAUT3, the RAIDUA model, which has emerged in the field of AI acceptance research in recent years, also provides a targeted perspective for understanding AI adoption intention. When exploring AI acceptance among female entrepreneurs in Pakistan, Rasheed et al. defined the RAIDUA model as a "multi-dimensional AI acceptance framework that integrates technology barriers[17], social support, ethical perception, and self-efficacy", with its core contributions manifested in three aspects: it first realizes the subdivision of technology barriers by breaking down the traditional "facilitating conditions" into "infrastructure barriers" and "skill barriers", and confirms that the negative effect of skill barriers on female entrepreneurs' AI adoption intention is significantly higher than that of infrastructure barriers; second, it embeds ethical perception, emphasizing that the "ethical compliance" of AI is a prerequisite factor affecting trust formation, and its effect intensity is particularly prominent in high-uncertainty scenarios such as entrepreneurial decision-making; third, it achieves the contextualization of social support, pointing out that in the context of collectivist culture, the driving effect of "community demonstration support" on adoption intention is higher than that of "personal achievement support" in individualistic culture.

Due to its unique tool attributes and increasingly prominent "quasi-subjectivity" characteristics[18], GAI has rendered the AI anxiety it induces unprecedentedly complex and multi-dimensional. Specifically, such anxiety may include: existential anxiety about AI replacing existing professional roles[19]; "digital divide" or AI sociotechnical blindness anxiety caused by the pace of technological iteration outpacing individual learning capabilities[20]; and philosophical anxiety regarding the loss of human cognitive sovereignty and blurred ethical boundaries in human-machine collaboration[21].These different types of AI anxiety do not exist in isolation or simply overlap, but interact through complex psychological mechanisms to influence individuals' evaluation of technical value and their final usage decisions. For instance, AI job displacement anxiety may negatively affect performance expectancy by reducing users' trust in the effectiveness of AI systems[22]; in contrast, AI sociotechnical blindness anxiety may stimulate individuals' adaptive learning motivation, thereby enhancing their effort expectancy[23]. This "double-edged sword" effect significantly challenges the single-path assumption in traditional anxiety research that "anxiety necessarily inhibits behavior", highlighting the urgent need to construct a more refined and dynamic theoretical framework to explain its inherent logic.

The higher education context provides a unique and crucial field for in-depth exploration of this issue. As the core force in knowledge innovation and talent cultivation, faculty and students in higher education institutions are not only active explorers and potential high-frequency users of GAI but also the group most sensitive to technical ethics, academic integrity, and educational equity[24]. Their inherent disciplinary background differences may shape distinct technical cognitive paradigms and value judgment standards[25], thereby influencing their acceptance attitudes towards GAI and the intensity of anxiety perception. Meanwhile, individuals' AI self-efficacy serves as a key psychological resource that may play an important moderating role between AI anxiety and GAI adoption intention, buffering or exacerbating the negative impacts of anxiety[26]. Although existing studies have begun to focus on the independent impacts of disciplinary differences or self-efficacy, few have systematically examined how these variables interact in the transmission path of AI anxiety. This has led to insufficient academic understanding of practical and theoretical questions such as "which groups are more

vulnerable to specific types of AI anxiety" and "under what conditions can anxiety be transformed into positive behaviors that promote technology adoption", resulting in obvious research gaps.

As in the Fig. 1, this study aims to construct an extended theoretical analysis framework encompassing multi- dimensional dimensions of AI anxiety, mediating transmission mechanisms, and boundary moderating variables by organically integrating the core elements of the TAM and UTAUT models[27], in order to more accurately reveal the complex formation mechanism of GAI adoption intention among the higher education community[28]. Specifically, this study will focus on three differentiated AI anxiety constructs: AI learning anxiety, AI sociotechnical blindness anxiety, and AI job displacement anxiety. It will systematically test how these anxiety dimensions indirectly influence individuals' GAI adoption intention through core cognitive variables in TAM/UTAUT. Additionally, this study will thoroughly investigate the moderating roles of disciplinary background and AI self-efficacy in the aforementioned anxiety-cognition-intention transmission path.

The theoretical and practical significance of this study is reflected in the following aspects.Theoretically, this study seeks to address the deficiencies of existing technology acceptance models in the research on emotional dimensions. By deconstructing the multi-dimensional structure of AI anxiety and its differentiated impact mechanisms, it enriches and expands the emotional explanatory perspective of technology acceptance theories, providing a new theoretical lens and empirical basis for understanding the "emotion-cognition-behavior" transmission chain of emerging intelligent technologies. Practically, the findings of this study will offer scientific empirical support for higher education institutions to formulate differentiated and targeted GAI promotion strategies and training programs for faculty and students. It will assist universities in effectively identifying and breaking down the "anxiety barriers" in the process of technology diffusion, thereby promoting the responsible and efficient application of GAI in higher education, and ultimately serving the improvement of educational quality and the cultivation of innovative talents. Therefore, the core scientific questions addressed in this study include: (1) How do different types of AI anxiety differentially influence the GAI adoption intention of the higher education community? (2) What are their specific impact paths and inherent mediating mechanisms (especially through TAM/UTAUT core variables)? (3) What boundary conditions or interactive moderating roles do disciplinary background and AI self-efficacy play in the aforementioned impact processes?

To more clearly present the research gaps, Table 1 summarizes the core studies in the field of technology acceptance and the educational applications of GAI, and clarifies the differences and extensions between this study and previous research. Based on the latest research progress integrating UTAUT3[29] and the RAIDUA model (Revised Artificial Intelligence Device Use Acceptance Model)[30], the core gaps in existing research can be summarized into three aspects, which also constitute the core innovative starting point of this study. Firstly, the

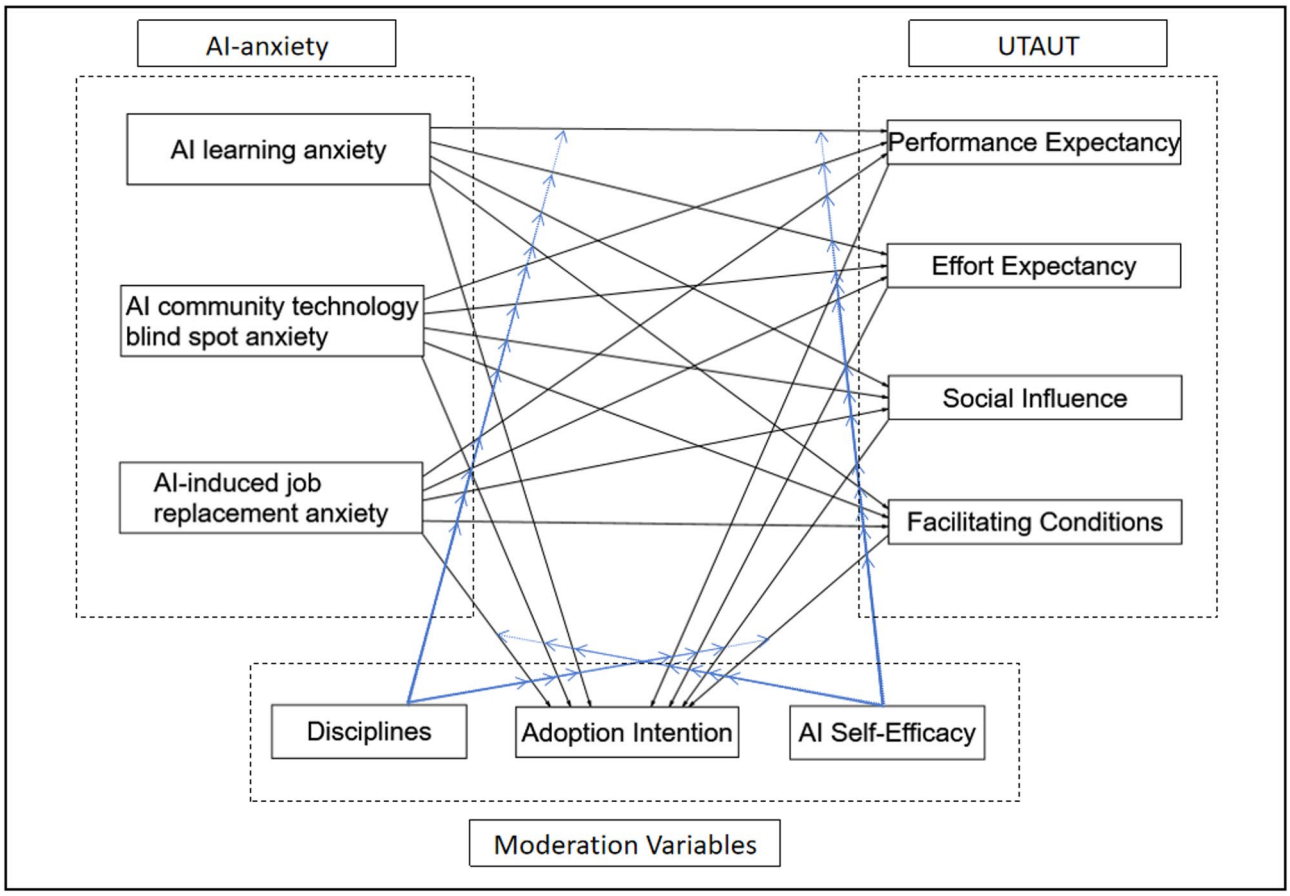

Fig. 1. The conceptual model.

Authors & Year	Research Topic	Core Opinions / Research Findings	Research Methods	Differences and Extensions of This Study
Venkatesh et al., 2003	Construction of the UTAUT Theory.	Identified four core variables influencing technology adoption intention, and incorporated moderating variables such as gender.	Empirical research; integrating multiple technology acceptance models.	1.Expanded the UTAUT framework by adding a multi-dimensional AI anxiety construct as an antecedent variable; 2. Focused on the GAI context in higher education and supplemented two targeted moderating variables: disciplinary background and AI self-efficacy.
Davis, 1989	Construction of the TAM Model	Performance expectancy and perceived ease of use are key factors affecting technology acceptance.	Empirical research; based on the Theory of Reasoned Action.	1.Integrated the TAM and UTAUT frameworks to enhance the model's explanatory power; 2.Broke through the limitations of rational factors and systematically analyzed the mechanism of action of emotional factors (AI anxiety).
Zhang et al., 2022	Anxiety Factors in Educational Technology Adoption.	Technology anxiety has a significant negative impact on educational technology adoption intention.	Questionnaire survey; structural equation modeling.	1.Refined the dimensions of AI anxiety, distinguishing between learning anxiety, community technical blindness anxiety, and job displacement anxiety; 2. Revealed the differentiated impact paths of different types of anxiety, breaking through the cognition of a single negative effect.
Li et al., 2023	University Teachers' AI Technology Adoption Intention.	Self-efficacy positively affects AI technology adoption intention, while technology complexity negatively affects adoption intention.	Questionnaire survey; regression analysis.	1. Constructed a complete "anxiety- mediation- moderation" theoretical framework instead of single-variable analysis; 2. Included both faculty and student groups, expanding the research object scope and improving the applicability of conclusions.
Wang et al., 2024	Adaptability of UTAUT2 in the Educational Application of GAI.	Performance expectancy and social influence are the main driving factors for GAI adoption.	Questionnaire survey; PLS-SEM analysis.	1.Introduced multi-dimensional AI anxiety variables to enrich the research on antecedent influencing factors; 2.Examined the moderating effect of disciplinary background, revealing the adoption difference mechanism among different disciplinary groups.
Gupta et al., 2023	E-leadership and virtual communication adoption by educators (UTAUT3)	Individual innovation and social influence interact to explain adoption intention; technology anxiety weakens effort expectancy.	Questionnaire survey; PLS-SEM	(1) Further subdivides anxiety into 3 dimensions (learning, sociotechnical blindness, job displacement); (2) Examines disciplinary differences not covered in their study.
Salam et al., 2025	MOOC utilization in higher education (UTAUT3)	Trust in technology content accuracy mediates performance expectancy and adoption.	Questionnaire survey; structural equation modeling	(1) Introduces AI self-efficacy to regulate trust-anxiety interaction; (2) Focuses on GAI instead of MOOCs, addressing academic originality concerns.
Ishengoma & Shao, 2025	AIoT integration in library systems (UTAUT3)	Risk perception (data privacy) negatively affects adoption, mitigated by "trust-habit" chain mediation.	Case study; survey	(1) Links risk perception to specific anxiety types (e.g., job displacement anxiety); (2) Explores academic scenario instead of library systems.
Rasheed et al., 2025	AI acceptance by female entrepreneurs (Application of RAIDUA Model)	Applied the RAIDUA Model and found that skill barriers (β= -0.32) have stronger negative effects than infrastructure barriers; ethical perception drives trust in AI adoption.	Interview; survey	(1) Adapts the RAIDUA Model's "skill barrier" construct to AI learning anxiety in higher education contexts, extending it from a functional barrier to an emotional anxiety dimension; (2) Supplements disciplinary background as a moderator not included in the original RAIDUA Model; (3) Integrates RAIDUA's AI-specific barrier logic with UTAUT3's emotional and individual innovation constructs, addressing the original model's neglect of multi-dimensional anxiety.
Zhang et al., 2022	Anxiety in educational technology adoption	Technology anxiety has a single negative effect on adoption intention.	Questionnaire survey; structural equation modeling	(1) Reveals "double-edged sword" effect of sociotechnical blindness anxiety; (2) Integrates UTAUT3 constructs to avoid single-variable analysis.

Table 1. Literature review of related studies and the differences and extensions of this Study.

fragmentation of emotional variable integration. As a mainstream framework in technology acceptance research, UTAUT3 has significantly improved the explanatory power for intelligent technologies by integrating constructs such as individual innovativeness, perceived trust, and perceived risk. However, it has not systematically decomposed the multidimensional attributes of "anxiety", nor has it verified the differential impacts of different anxiety types on its core constructs. In contrast, the RAIDUA model, as a revised framework specifically targeting AI acceptance[30], has expanded AI-specific constructs and subdivided technological barriers based on the multi-construct logic of UTAUT3. Yet, it completely ignores emotional variables, making it difficult to explain the "anxiety-defense" responses triggered by highly uncertain technologies like GAI.Secondly, insufficient adaptability to the higher education context. Although studies by Salam et al. on MOOCs and Gupta et al. on virtual communication are based on educational scenarios[11,13], they have not designed variables targeting the "academic attributes" of GAI. The original application scenario of the RAIDUA model focuses on AI services in the hotel industry, while Rasheed et al. have only extended it to the female entrepreneur group[17]. Neither has examined the paradigmatic impact of key disciplinary background differences in higher education on technology cognition — a difference that is precisely the core source of heterogeneity in technology acceptance in academic scenarios. Thirdly, insufficient depth of moderating mechanisms. Existing studies either focus on regional resource differences[16,17] or single moderating variables[15]. They have not yet constructed a "contextual variable + individual variable" dual-moderation framework, making it difficult to reveal the heterogeneous mechanisms of anxiety transmission among different groups. Addressing the aforementioned gaps, this study organically integrates the rational constructs of UTAUT3 with the subdivided logic of AI technological barriers from the RAIDUA model. By embedding multidimensional AI anxiety and dual moderating variables, it constructs an extended framework more adaptable to the GAI context in higher education, thereby filling the triple gaps in existing research regarding "emotional integration, contextual adaptability, and moderating depth".

Theoretical foundations

Unified theory of acceptance and use of technology 1 and 3 (UTAUT1, UTAUT3)

The Unified Theory of Acceptance and Use of Technology (UTAUT) was proposed by Venkatesh et al. in 2003, with the core objective of integrating fragmented technology acceptance models to construct a comprehensive and explanatory theoretical framework[10]. This theory identifies four key factors influencing users' technology adoption intention and usage behavior: performance expectancy, effort expectancy, social influence, and facilitating conditions. These core constructs can explain 70% of the variance in technology adoption behavior[31], significantly outperforming earlier single models. Meanwhile, it incorporates moderating variables such as gender and age, clarifying the moderating effects of individual and situational factors, and enhancing the model's universality and predictive validity.

In the field of higher education, the UTAUT framework is widely used to analyze the adoption mechanisms of educational technology tools[32,33]. However, it has limitations when applied to emerging technologies with high intelligence, high autonomy, and ethical complexity such as GAI. To address this, relevant studies have expanded the framework while retaining UTAUT's core structure, incorporating multi-dimensional AI anxiety constructs as antecedent driving variables. They analyze the indirect impact of these constructs on adoption intention through UTAUT's core paths. This integrated framework not only preserves the advantage of analyzing rational paths but also supplements the emotional dimension, providing a dynamic perspective for understanding the "emotion-cognition-behavior" transmission chain of GAI in higher education.

As the latest achievement in the UTAUT series, UTAUT3[29] has added four core constructs— habit, perceived trust, perceived risk, and individual innovativeness—on the basis of UTAUT2. Its multi- construct inclusiveness has laid an important theoretical foundation for the development of subsequent AI acceptance models. For instance, the RAIDUA model[30] has further integrated AI-specific constructs and subdivided technological barriers based on the logical framework of UTAUT3, forming an acceptance model more adaptable to AI human-computer interaction scenarios. Among these constructs, the individual innovativeness in UTAUT3 provides key supplementary value for explaining the early adoption behavior of emerging intelligent technologies, and is particularly suitable for complex and disruptive technological contexts such as GAI in higher education.

The operational definition of individual innovativeness in UTAUT3 is "an individual's personality trait and behavioral intention to proactively try new things and explore new technologies", and its core mechanism of action is reflected in three aspects. Firstly, the antecedent driving effect: individual innovativeness positively influences performance expectancy and effort expectancy by reducing technological cognitive barriers. Individuals with high innovativeness are more likely to perceive the potential value of technology and adopt a more inclusive attitude toward the costs of learning new technologies. Farooq et al. have empirically verified that in the context of educational technology, individual innovativeness has a significant impact on the path coefficients of performance expectancy and effort expectancy[29], and this driving effect is more pronounced in scenarios with high technological complexity—consistent with the technical characteristics of GAI in this study.

Notably, although the RAIDUA model does not directly incorporate individual innovativeness, its "skill barriers" construct has an inherent logical connection with the effort expectancy and individual innovativeness of UTAUT3. Individuals with high individual innovativeness are more likely to overcome skill barriers through proactive exploration. This study further expands the RAIDUA's "skill barriers" into emotional "AI learning anxiety", exploring the buffering effect of individual innovativeness in the process where anxiety is transformed into motivation.

In the application of GAI in higher education, the personal innovativeness construct of UTAUT3 can also address the limitations of the existing framework in this study. Although the existing model includes "AI self-efficacy" as a moderating variable, "AI self-efficacy" focuses on "ability belief" while personal innovativeness focuses on "behavioral tendency", and the two are complementary — for instance, university teachers and students with high AI self-efficacy but low personal innovativeness may "have the ability to use GAI but be unwilling to try it", while high personal innovativeness can promote the transformation of their ability into actual adoption behavior. This supplement not only improves the theoretical logic of the "individual dimension" in technology acceptance but also provides a direction for optimizing the moderating mechanism in subsequent studies.

In recent years, research on the application of UTAUT in intelligent technology contexts has continued to deepen[34,35]. Henseler et al.'s PLS-SEM analysis guidelines provide a standardized method for empirical testing of models[36], while Venkatesh et al.'s research highlights the importance of emotional factors in emerging technology contexts[37]. Gerlich et al.'s study on the educational application of ChatGPT found that the effect intensity of traditional UTAUT core variables has changed, with the driving role of social influence significantly enhanced[38]. This echoes relevant empirical results and provides theoretical support for integrating emotional variables to expand the model.

The exclusion of constructs such as hedonic motivation and habit from UTAUT2 in existing studies is mainly based on two considerations: firstly, the research objects focus on faculty and students in higher education, whose adoption intention is centered on functional needs for teaching and research, so hedonic motivation is not a major driving factor, which is essentially different from the consumer technology usage scenarios applicable to UTAUT2. Secondly, GAI is still in the early stage of application in higher education[39], and most faculty and students have not formed stable usage habits, resulting in insufficient behavioral basis for measuring the habit construct.Therefore, relevant studies focus on UTAUT's core constructs and incorporate emotional and moderating variables in combination with the research context, which is more in line with the actual needs of current research. The theoretical basis for integrating TAM and UTAUT to construct an extended framework lies in three aspects: firstly, the concise explanatory power of TAM and the wide coverage of UTAUT complement each other; secondly, the PLS-SEM method is suitable for empirical testing of complex models;

thirdly, it supplements the research gap of emotional factors in traditional models, adapting to the unique needs of GAI's application in higher education.

Artificial intelligence anxiety

Artificial Intelligence anxiety consists of three dimensions: AI learning anxiety, AI sociotechnical blindness anxiety, and AI job displacement anxiety. Among them, the core of AI learning anxiety is individual cognitive anxiety of "incompetence", caused by the complexity and learning costs of the technology itself, with emotions directed at self-ability negation. Based on the perception of "ability gap" in technology diffusion theory, its impact path mainly directly inhibits effort expectancy and performance expectancy, and is significantly moderated by individual self-efficacy. The core of AI sociotechnical blindness anxiety is social comparison anxiety of "social marginalization", arising from the pressure of technical competence evaluation in academic communities, with emotions directed at the loss of social identity. Based on social identity theory and academic capital competition theory, its impact path presents a "double-edged sword" characteristic. The core of AI job displacement anxiety is existential anxiety of "loss of professional value", caused by GAI's substitution threat to core academic functions, with emotions directed at the crisis of professional identity. Based on technology substitution theory and professional identity theory, its impact path mainly comprehensively inhibits the core variables of technology acceptance, and has the most significant and direct inhibitory effect on adoption intention. Details are shown in Table 2.

AI anxiety

AI learning anxiety

AI learning anxiety specifically refers to the cognitive and emotional stress endured by the higher education group when encountering and mastering GAI technology[40]. It stems primarily from concerns about technological complexity, a steep learning curve, and self-perceived learning capabilities[41], which is highly consistent with the "skill barriers" construct proposed by the RAIDUA model[30]. The latter defines "skill barriers" as adoption obstacles arising from an individual's lack of professional skills required to use GAI. This study further expands it from a functional barrier to an emotional perception of anxiety, exploring the psychological transformation mechanism from "objective capability gaps" to "subjective emotional anxiety". From the perspective of the technology diffusion theory[42], such anxiety reflects an individual's perception of "competence gaps" in the process of acquiring new skills. This perception becomes more pronounced especially when the pace of technological iteration exceeds the speed of personal cognitive updating.

Unlike traditional computer anxiety[43], which involves generalized fear of basic operational skills[44], AI learning anxiety exhibits distinct domain specificity—it focuses on the technical black-box characteristics unique to GAI, such as natural language processing and machine learning principles, as well as the resulting "algorithm comprehension gap[45]. " The higher education context amplifies the uniqueness of AI anxiety[46]. On one hand, teachers and students, as knowledge transmitters and creators, face professional pressure to integrate AI tools into teaching and research paradigms; on the other hand, discrepancies in the compatibility between disciplinary knowledge systems and AI technological logic may exacerbate anticipated learning barriers.

Gerlich et al.'s empirical research based on ChatGPT application scenarios found that AI learning anxiety among the higher education community exhibits the characteristic of "competence cognitive bias"[47]—that is, individuals' perception of the complexity of GAI is often higher than the actual learning difficulty, and this bias is more pronounced in the humanities and social sciences groups. The study further pointed out that targeted technology decomposition training can effectively reduce this cognitive bias, thereby alleviating learning anxiety. This provides important empirical support for understanding the formation mechanism and intervention paths of AI learning anxiety.

AI sociotechnical blindness anxiety

AI sociotechnical blindness anxiety denotes the deep-seated fear that individuals may be marginalized within their academic or professional communities due to insufficient technical competence or outdated knowledge[48]. Its essence lies in digital existential anxiety rooted in social comparison and academic capital competition[49].

Feature Dimensions	AI Learning Anxiety	AI sociotechnical blindness anxiety	AI Job Displacement Anxiety
Core Concept	Cognitive pressure from insufficient technical learning capabilities.	Social pressure of being marginalized in the community due to technological backwardness.	Existential pressure from the replacement of core professional functions.
Core Causes	Technological complexity, steep learning curve, insufficient self-directed learning abilities.	Changes in technical evaluation standards in academic communities, social comparison pressure.	Disruptive potential of GAI on the core functions of teaching and scientific research.
Emotional Orientation	Self-competence denial, learning frustration.	Social identity devaluation, fear of loss of academic capital.	Loss of professional value, identity crisis.
Theoretical Foundations	Technology Diffusion Theory, Cognitive Load Theory.	Social Identity Theory, Academic Capital Theory.	Technology Replacement Theory, Professional Identity Theory.
Main Mechanisms of Action	Negatively affects effort expectancy and performance expectancy.	Positively affects effort expectancy and adoption intention; negatively affects performance expectancy.	Comprehensively negatively affects performance expectancy, effort expectancy, social influence, etc.
Key Moderating Variables.	AI Self-Efficacy.	Disciplinary Background, Community Support Network.	Professional Role Cognition, Intensity of Perceived Technological Replacement.

Table 2. Comparison of core characteristics and mechanisms of action among three types of AI Anxiety.

Within higher education settings, the sources of this anxiety are distinctly socially constructed and context-dependent. Firstly, the rapid iteration of GAI tools and their penetration into academic applications have reshaped the technical competency evaluation criteria of academic communities, rendering "technological blind spots" potential markers of deprivation of knowledge discourse power. Secondly, universities, as high-density knowledge interaction networks, feature online academic communities and offline peer review mechanisms that form a "panoptic gaze field" of technological visibility, significantly amplifying individuals' sense of shame regarding "technological backwardness" under communal scrutiny[50].

The core characteristic of AI sociotechnical blindness anxiety lies in its social-driven — its emergence stems not only from technology itself but more from expectations that technological gaps may lead to devaluation of academic identity and loss of social capital[51]. This anxiety may trigger two behavioral mechanisms: adaptive catch-up behavior[52], and value negation defense[53]. This study incorporates this construct into its framework precisely to deconstruct this "fear-motivation" paradox and reveal how social pressure influences final adoption decisions by altering technological cognition and behavioral strategies.

AI job displacement anxiety

AI job displacement anxiety specifically refers to the profound concern among higher education practitioners that GAI may replace their core professional functions[54]. This anxiety arises from GAI's disruptive potential in knowledge production and dissemination, such as automated generation of teaching plans, homework grading, research paper writing, and even curriculum design, leading teachers and students to worry that their teaching value, academic contributions, and professional identities face fundamental challenges[55]. Theoretically, technology displacement theory and career identity theory collectively illuminate the formation mechanism of this anxiety. The uniqueness of the higher education context exacerbates this effect: academic work has long relied on "exclusively human capabilities" such as critical thinking and knowledge innovation, yet GAI's breakthroughs in complex reasoning and creative output have blurred the human-machine capability boundary[56], prompting practitioners to reevaluate their positioning within the academic ecosystem.

When individuals perceive GAI as a "professional threatener" rather than an "ability enhancer," their cognitive appraisal undergoes systematic bias. They may question the potential of GAI to improve work effectiveness, avoid learning complex GAI tools, and downplay the influence of technical resource support and social normative pressure on their behavioral decisions[57]. This "defensive cognition" renders AI job displacement anxiety a significant psychological barrier to GAI diffusion in higher education, with an intensity far exceeding functional obstacles in traditional technology acceptance models, underscoring the theoretical necessity of incorporating the emotional-cognitive dimension in emerging technology adoption research.

The latest research by Gerlich and colleagues reveals that the intensity of AI job displacement anxiety is closely associated with individuals' perception of the "human-AI collaboration boundary"[58]. When faculty and students in higher education institutions have a clear understanding of GAI's role as an auxiliary tool, their displacement anxiety is significantly reduced; in contrast, when the application scenarios of the technology blur the boundary between the core functions of humans and GAI, the anxiety level rises sharply. This finding provides a crucial theoretical supplement for this study's exploration of the transmission mechanism of AI job displacement anxiety, and also explains why there are significant differences in the perception of professional threats posed by GAI among different groups.

Moderating Variables—(Disciplines, AI Self-Efficacy)

Disciplines and AI self-efficacy serve as critical moderating variables within the psychological mechanism of GAI adoption intention. They shape individuals' cognitive appraisals of anxiety and coping resources, thereby influencing the pathway through which anxiety transforms into behavioral intentions.

Disciplines represents divergent cognitive paradigms and technological value systems. Groups within science, technology, engineering, and mathematics (STEM) disciplines typically operate within an instrumental rationality framework, focusing on the functional efficacy and feasibility of GAI[59]. Their anxiety often stems from adaptability pressures induced by rapid technological iteration. Conversely, groups within the humanities and social sciences often adopt a critical paradigm, vigilant against the erosion of academic subjectivity, ethical considerations, and traditional modes of knowledge production by technology[60]. Their anxiety is more readily associated with identity crises and the defense of disciplinary traditions. This divergence leads to different behavioral responses to similar anxiety triggers. STEM individuals may reframe learning anxiety as a surmountable challenge, potentially motivating skill acquisition, while humanities and social sciences individuals may interpret it as a threat to disciplinary values, reinforcing avoidance tendencies.

AI self-efficacy, functioning as an individual psychological resource, moderates the intensity of anxiety's impact on mediating variables related to technology acceptance. Individuals with high self-efficacy possess stronger beliefs in their ability to master technology, enabling them to cognitively reframe anxiety as a catalyst for capability development[61]. When individuals are confident in their capacity to utilize GAI, learning anxiety may paradoxically foster a more positive assessment of the technology's value[62]. This aligns with the "stress-to-motivation conversion model" posited by social cognitive theory — self-efficacy facilitates problem-focused coping strategies, effectively transforming anxiety from a "hindering factor" into an "action signal."

Hypothetical development

AI learning anxiety

AI learning anxiety emerges as a core emotional barrier within the diffusion of GAI technology in higher education contexts, exerting multifaceted inhibitory effects on the technology adoption intention of both educators and students[63]. Regarding effort expectancy, when educators or students experience AI learning anxiety due to concerns about technological complexity or perceived inadequacies in their learning capabilities,

they tend to overestimate the cognitive load and time investment required to master GAI tools[64]. This "perceived difficulty amplification effect" directly diminishes effort expectancy. Concerning performance expectancy, AI learning anxiety triggers doubts about the alignment between individual capabilities and task requirements, leading individuals to question whether they can effectively utilize GAI to enhance their work performance[65]. Such skepticism undermines performance expectancy — the belief that GAI will improve task outcomes. With respect to social influence, individuals grappling with frustration during technology skill acquisition may avoid experiential exchanges about GAI applications within their academic communities or even actively reject demonstrations of technology use by peers[66]. This "social avoidance" strategy diminishes the impact of social influence. Pertaining to facilitating conditions, individuals experiencing AI learning anxiety often fall into a state of "resource blindness"; even when institutions provide ample technical training resources, their attention remains fixated on perceived personal skill deficits, leading to an underestimation of the availability and effectiveness of external support[46]. This directly weakens perceptions of facilitating conditions.

Ultimately, the negative impact of AI learning anxiety on adoption intention stems from the cumulative inhibitory effects of these mediating pathways. When individuals persistently experience frustration during technology learning, it activates a psychological defense mechanism characterized by "anticipated failure-behavioral avoidance." Consequently, even when rationally acknowledging the value of the technology, individuals may delay or abandon usage decisions[67]. This suppression of intention is particularly pronounced in rapidly evolving technological environments. Synthesizing these arguments, we propose the following hypotheses:

H1a AI learning anxiety negatively influences effort expectancy;

H1b AI learning anxiety negatively influences performance expectancy;

H1c AI learning anxiety negatively influences social influence;

H1d AI learning anxiety negatively influences facilitating conditions;

H1e AI learning anxiety negatively influences adoption intention.

AI sociotechnical blindness anxiety

AI sociotechnical blindness anxiety, as a form of digital existential anxiety rooted in social comparison, exerts a complex cognitive-affective interactive influence on the core variables of technology acceptance[68]. Within the context of higher education, when individuals perceive their GAI technical capabilities as lagging behind the average level of the academic community, this threat perception of "technical marginalization" may trigger dual psychological defenses. On one hand, an overestimation of technological complexity directly undermines rational assessment of learning costs, fostering negative prediction regarding effort expectancy[69]. On the other hand, the motivational drive to maintain academic identity may prompt strategic devaluation of AI tools' value, whereby individuals protect their sense of self-worth by questioning the practical utility of such technologies in academic production — a defensive cognition that directly translates into negative impacts on performance expectancy[70].

AI sociotechnical blindness anxiety may attenuate the transmission of social influence on adoption intention[71] through two pathways: firstly, the "information echo chamber" effect, whereby anxious individuals tend to actively avoid technology discussion communities, resulting in biased information sampling for perceived social norms[72]; secondly, the "reverse identification" mechanism, where when technical competence becomes a marker of group stratification, individuals in technologically disadvantaged positions may reconstruct group boundaries by negating the technological values of mainstream groups, with such sub-cultural resistance directly digesting the driving force of social influence. In terms of facilitating conditions, AI sociotechnical blindness anxiety intensifies feelings of distrust toward technological environmental support[73], and this selective attention bias leads to evaluations of facilitating conditions that are significantly lower than objective levels. Ultimately, these multi-path negative influences converge to systematically inhibit adoption intention. In summary, we propose the following hypotheses:

H2a AI sociotechnical blindness anxiety negatively affects effort expectancy;

H2b AI sociotechnical blindness anxiety negatively affects performance expectancy;

H2c AI sociotechnical blindness anxiety negatively affects social influence;

H2d AI sociotechnical blindness anxiety negatively affects facilitating conditions;

H2e AI sociotechnical blindness anxiety negatively affects adoption intention.

AI job displacement anxiety

Within the context of GAI applications in higher education, AI job displacement anxiety constitutes a distinct psychological barrier[57]. This anxiety originates from a profound fear that AI technology may erode the core value of professions, culminating in a defensive cognitive framework. Grounded in technology displacement theory and occupational identity theory, this anxiety inhibits the formation of technology acceptance mechanisms through multiple pathways.

Regarding performance expectancy, when individuals perceive GAI as a threat to their profession rather than a tool, they systematically devalue its functional utility[74]. For instance, instructors concerned that AI-generated lesson plans might undermine their pedagogical expertise may tend to deny GAI's potential for enhancing classroom efficiency or enabling personalized tutoring[75]. This cognitive bias directly diminishes performance expectancy[76]. Concerning effort expectancy, individuals experiencing high levels of AI job displacement anxiety often perceive acquiring GAI skills as reinforcing "professional betrayal" rather than recognizing it as an opportunity for capability enhancement[77]. This psychological resistance amplifies the perceived difficulty of learning the technology; even when confronted with user-friendly interfaces or structured training programs, individuals may subjectively assess the learning costs as prohibitively high, thereby weakening effort expectancy[78]. Pertaining to social influence, although the academic community may generate normative pressure favoring GAI adoption, anxious individuals tend to employ a "value negation" strategy to safeguard professional self-esteem[79]. For example, they may interpret peers' use of GAI tools as "academic opportunism" or "intellectual indolence," rather than innovative practice. This cognitive reframing attenuates the impact of social influence on behavioral intention. With respect to facilitating conditions, when individuals question the compatibility of GAI with their career development, even when institutions provide ample technical support, they may perceive these resources as "catalysts" accelerating professional obsolescence rather than empowering tools[26]. This goal conflict impedes the translation of facilitating conditions into behavioral enablers.

Ultimately, AI job displacement anxiety, via this chain of inhibitory mediating mechanisms, leads individuals to avoid engagement with GAI tools, creating a self-reinforcing "anxiety-cognitive devaluation- behavioral avoidance" loop. Breakthroughs in GAI within this domain exacerbate perceived threats to professional autonomy, transforming AI anxiety from merely an emotional reaction into a structural impediment to technology diffusion[73]. Therefore, the following hypotheses are proposed:

H3a AI job displacement anxiety negatively influences effort expectancy.

H3b AI job displacement anxiety negatively influences performance expectancy.

H3c AI job displacement anxiety negatively influences social influence.

H3d AI job displacement anxiety negatively influences facilitating conditions.

H3e AI job displacement anxiety negatively influences adoption intention.

Performance expectancy

Performance expectancy reflects an individual's subjective belief that using a specific technology will enhance their job or learning performance. In the higher education context, the performance expectancy of GAI tools manifests in the perceived value by faculty and students regarding their potential to improve teaching efficiency, support research innovation, and empower personalized learning[80]. According to the core tenets of the UTAUT, when users anticipate that a technology will significantly improve task performance, their adoption intention is strengthened[81]. The unique advantages of GAI establish a clear expectation of performance improvement for the higher education community. This expectation stems not only from the tools' functional features but is also rooted in the perpetual pursuit of efficiency and innovation inherent in higher education scenarios. Faculty expect GAI to alleviate the burden of repetitive tasks, thereby allowing them to focus on higher-order instructional design; students, meanwhile, anticipate that GAI tools will help them overcome cognitive bottlenecks and achieve deep learning and knowledge construction. Previous technology acceptance research has confirmed the universal driving effect of performance expectancy on adoption intention[82,83], and GAI's breakthrough performance in complex cognitive tasks further reinforces the theoretical relevance of this pathway. Therefore, we propose the hypothesis:

H4 Performance expectancy positively influences adoption intention.

Effort expectancy

Effort expectancy reflects an individual's subjective assessment of the cognitive and behavioral costs required to use a specific technology[84]. In the context of GAI, the effort expectancy of the higher education community encompasses not only judgments about the technical operation's complexity but also involves deeper considerations regarding cognitive adaptability within human-AI collaboration models. The "black-box nature" of GAI tools and the high specialization of academic tasks mean that users' perception of the learning cost directly influences their adoption decisions. Based on the core assumptions of UTAUT[85], when individuals expect that the process of using the technology will not excessively consume cognitive resources or create significant learning barriers, their adoption intention is strengthened. The higher education setting amplifies this effect. Faculty and students face dual pressures from teaching and research, and their limited time resources make them more sensitive to the "usability threshold" of technology. Prior research indicates that in educational technology diffusion, effort expectancy promotes tool integration by reducing "psychological resistance costs[86]," particularly when there is a significant logical gap between the technology and the disciplinary knowledge system; low effort expectancy can effectively mitigate cognitive conflicts and enhance adoption intention[82]. Therefore, we propose the hypothesis:

H5 Effort expectancy positively influences adoption intention.

Social influence

Social influence in the higher education context shapes individuals' GAI adoption intention through the normative pressures of the academic community and mechanisms of social identity[87]. UTAUT posits that social influence reflects the intensity of an individual's perception that significant others expect them to use the technology[88]. This expectation operates through three distinct pathways. Firstly, the technological usage norms within academic communities constitute implicit "professional pressure," compelling individuals to adopt mainstream technological tools to maintain academic capital and collaborative opportunities. Secondly, peers' successful application cases of GAI provide vicarious experience, reducing perceptions of uncertainty surrounding technology use. Thirdly, policy directives and resource support from institutional administrators further reinforce the social legitimacy of technology adoption. As a high-density knowledge community, the academic evaluation systems and collaborative networks of the higher education population are highly dependent on social interaction, making the permeation of social influence on adoption intention far stronger than in general organizational contexts[89]. When faculty and students perceive that GAI has become the "lingua franca" of academic discourse or an "essential tool" for research innovation, the need for social identity will significantly enhance their adoption intention. Therefore, we propose the hypothesis:

H6 Social influence positively influences adoption intention.

Facilitating conditions

Facilitating conditions reflect an individual's perception of the extent to which organizational or technological environments support their use of AI tools, encompassing objective factors such as resource accessibility, technological infrastructure adequacy, and training support[90]. Within the context of higher education, facilitating conditions directly influence adoption decisions by reducing external barriers to technology use[91]. When universities provide sufficient access to AI tools, technical support services, and customized training programs, teachers and students can more seamlessly integrate AI technologies into teaching and research processes, mitigating usage resistance stemming from resource inadequacy or lack of support. The UTAUT framework emphasizes the critical role of facilitating conditions in the early stages of technology diffusion[92], particularly for emerging technologies like GAI that demand continuous learning and environmental adaptation. Organizational resource investment and institutional safeguards thus constitute an important material foundation for adoption intention. From the perspective of environmental support in technology acceptance research, facilitating conditions enhance perceptions of use feasibility by eliminating external constraints, thereby positively promoting adoption intention. Based on the above, we propose the following hypothesis:

H7 Facilitating conditions positively influence adoption intention.

Moderating variables (Disciplines, AI Self-Efficacy)

Disciplines and AI self-efficacy, as key situational and individual variables, may systematically moderate the strength of the pathways between AI anxiety and technology acceptance mediators[93,94]. Disciplinary differences shape technological cognitive paradigms: STEM disciplines, long immersed in instrumental rationality, may see their technological anxiety more readily translated into pragmatic evaluations of efficacy improvement or weaken the negative effect of AI Sociotechnical blindness anxiety on performance expectancy. In contrast, humanities and social sciences groups, with their sensitivity to technological ethics and cognitive sovereignty, may intensify the inhibitory effect of job displacement anxiety on social influence. As a core psychological resource, AI self-efficacy equips individuals with stronger anxiety transformation capabilities — high self-efficacy individuals may amplify the positive driving effect of AI learning anxiety on performance expectancy while buffering the irrational arousal of AI Sociotechnical blindness anxiety on effort expectancy. Their interactive effect manifests in that the technological culture of STEM fields may inherently foster high self-efficacy, whereas humanities and social sciences groups, if enhancing efficacy through specialized training, can weaken disciplinary barriers in anxiety transmission. This dual-dimensional moderating mechanism suggests that the impact of anxiety on technology acceptance is not a static effect but rather a result of dynamic coupling between situational cognition and psychological resources. Based on the above, we propose the following hypothesis:

H8 Moderating variables partially moderate the relationships between independent variables and mediating variables.

Measurement tools and data collection

Questionnaire design

This study employs a structured questionnaire as the data collection instrument, which is constructed based on the UTAUT and theories related to AI anxiety. The questionnaire consists of three core sections: the first section provides research explanations, including the research purpose, implementation procedures, and an informed consent statement; the second section gathers demographic information, encompassing basic variables such as gender, ethnicity, and academic discipline; the third section comprises measurement indicators for the latent variables in the research framework, with a total of 35 items designed to assess latent variables including AI learning anxiety (4 items), AI Sociotechnical blindness anxiety (3 items), AI job displacement anxiety (3 items), performance expectancy (3 items), effort expectancy (4 items), social influence (4 items), facilitating conditions (4 items), AI self-efficacy (6 items), and adoption intention (4 items). All measurement items are quantified using a 7-point Likert scale (1 = Strongly Disagree, 7 = Strongly Agree), enabling comparable measurement of the influencing factors of AI anxiety on the adoption intention of GAI among university teachers and students through a standardized scoring system.

Data collection

This study employed a questionnaire survey method, selecting three universities in Sichuan Province (Xihua University, Chengdu University of Information Technology, and Chengdu Aeronautic Polytechnic) as the source of research samples. The research team constructed a standardized survey instrument using a professional online survey platform (Wenjuanxing, https://www.wjx.cn/) and implemented full-process quality control. Prior to questionnaire completion, researchers provided respondents with detailed explanations of the questionnaire structure, content analysis, and filling guidelines through the online platform, and ensured consistency of understanding via real-time online Q&A, strictly adhering to the principle of non-directionality throughout the process. A total of 500 questionnaires were distributed, and after data cleaning, 407 valid samples were obtained, resulting in an effective recovery rate of 81%. In accordance with the modeling specifications for structural equation modeling[95], this study determined the minimum sample size to be 190 based on the 10-times rule of the number of latent variable paths. The actual sample size significantly exceeds this threshold, providing sufficient statistical power for model parameter estimation. Table 3 summarizes the demographic characteristics of the participants, which offer valuable contextual references for interpreting the research results.

Control of common method bias

Given that the data of this study were collected through a single-source questionnaire survey, there may be a risk of common method bias. To ensure the reliability of the research results, this study implemented systematic controls from two aspects: procedural design and statistical testing. At the procedural level, an anonymous filling mode was adopted to protect the privacy of respondents, avoiding their ingratiating responses due to concerns; the measurement items of different latent variables were randomly ordered to break the cognitive association of dimensional relevance among items; the research purpose and filling norms were clearly stated in the questionnaire instructions to guide respondents to fill in based on their true perceptions. At the statistical level, Harman's single-factor test was used for post-hoc verification. All 35 measurement items were included in the unrotated exploratory factor analysis. The results showed that a total of 9 common factors with eigenvalues greater than 1 were extracted, and the variance explained by the first common factor was 28.63%, which did not reach the critical threshold of 40%[96]. This indicates that the common method bias in this study is within an acceptable range and will not significantly interfere with the research conclusions.

Results

This study utilized SmartPLS 4.0 for partial least squares structural equation modeling analysis, following the methodological framework proposed by Hair et al.[97]. For the research model containing 9 latent constructs and 35 measurement indicators, PLS-SEM was established as the core analytical method due to its advantages in handling complex causal networks and non-normally distributed data[98]. This method is particularly suitable for multivariate exploratory analysis in the theoretical development stage, providing a robust statistical basis for subsequent hypothesis testing[99].

Measurement model assessment

In structural equation modeling analysis, measurement model assessment is a critical step to ensure that the relationships between latent variables and observed variables are reasonable and theoretically consistent. This study conducted empirical analysis using partial least squares structural equation modeling, a method especially suitable for handling complex models and small sample data, effectively controlling measurement errors to verify theoretical hypotheses. As Fig. 2, the assessment of the measurement model in this study focused on three dimensions: For reliability assessment, internal consistency indicators (e.g., Cronbach's α, composite reliability [CR]) were used to evaluate the stability of observed variables in measuring latent variables; validity assessment included convergent validity (measured by indicators such as factor loadings and average variance extracted [AVE]) and content validity to ensure that observed variables fully reflect the theoretical connotations of latent variables; discriminant validity was verified by comparing the correlation coefficients between latent variables and the square root of AVE values to confirm the independence of measurements of different latent variables. This systematic assessment process laid a solid measurement foundation for subsequent structural model analysis and enhanced the credibility of the research conclusions.

Reliability test

Reliability testing is a critical step to verify data reliability, primarily used to assess the stability and internal consistency of measurement instruments. This study employed a dual-indicator approach for testing, combining Composite Reliability (CR) and Cronbach's α coefficient. As shown in Table 4, the CR values of all latent variables exceeded 0.9 (ranging from 0.906 to 0.996). However, the CR value for "AI Sociotechnical blindness anxiety" was greater than 0.980 (0.994), indicating an issue of excessive redundancy. Therefore, we examined the original

Profiles	Characteristics	Numbers of respondents	Percentage
Gender	Male	168	41%
	Female	239	59%
Disciplines	Humanities and Social Sciences	190	47%
	Engineering Science	217	53%

Table 3. Respondents' characteristics.

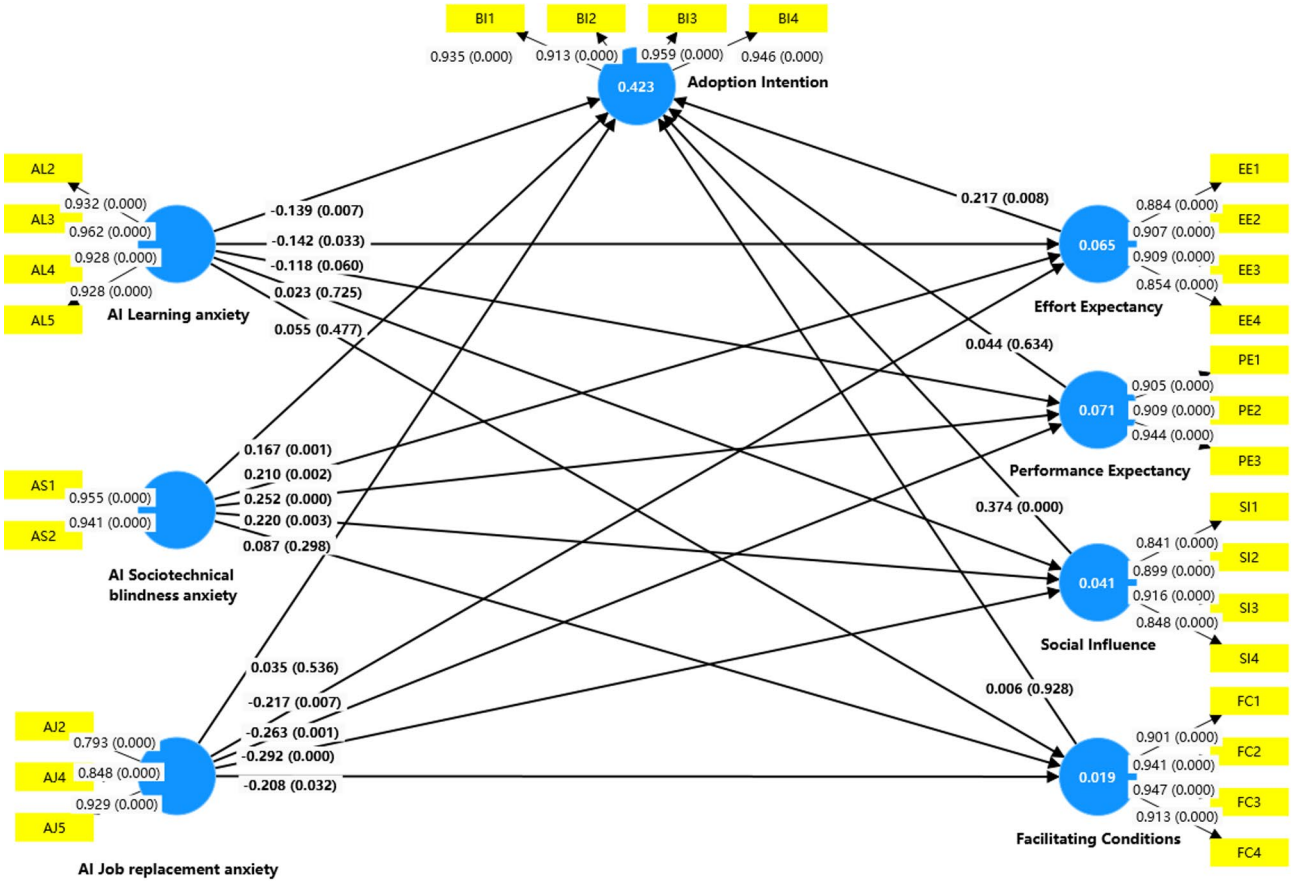

Fig. 2. The measurement model.

Constructs	Cronbach's alpha	CR	AVE
AI Self-Efficiency	0.902	0.904	0.673
AI Job replacement anxiety	0.825	0.905	0.737
AI Learning anxiety	0.954	0.958	0.879
AI Sociotechnical blindness anxiety	0.888	0.901	0.899
Adoption Intention	0.955	0.955	0.880
Effort Expectancy	0.911	0.913	0.790
Facilitating Conditions	0.944	0.946	0.857
Performance Expectancy	0.908	0.911	0.845
Social Influence	0.899	0.905	0.769

Table 4. Reliability and validity.

scale, which contained 3 items (e.g., "I worry about being marginalized by academic peers due to a lack of understanding of GAI", "I fear losing academic exchange opportunities because of backward AI technology", "I am anxious about being excluded from academic communities due to insufficient AI knowledge"). Due to the highly similar wording, the CR reached 0.996. After deleting the third item with the highest degree of repetition and retaining 2 items, the reliability was recalculated: CR = 0.900 (meeting the ideal range of $0.700 \leq CR \leq 0.950$), and Cronbach's $\alpha = 0.888$ (still satisfying the standard of > 0.8); these results indicate a high level of construct stability in the measurement model[90]. In addition, the Cronbach's α values of the latent variables all exceeded 0.8 (ranging from 0.825 to 0.955), further confirming good internal consistency among the scale items[100]. Taken together, these two indicators demonstrate that the measurement model meets the stringent reliability requirements of psychological and social science research, ensuring the reliability of measurement results with well-controlled errors.

Constructs	1	2	3	4	5	6	7	8	9
AI Self-Efficiency	**0.820**								
AI Job replacement anxiety	0.060	**0.859**							
AI Learning anxiety	-0.041	0.526	**0.938**						
AI Sociotechnical blindness anxiety	0.194	0.693	0.265	**0.948**					
Adoption Intention	0.688	-0.009	-0.156	0.169	**0.938**				
Effort Expectancy	0.599	-0.145	-0.199	0.022	0.578	**0.889**			
Facilitating Conditions	0.530	-0.119	-0.030	-0.044	0.447	0.696	**0.926**		
Performance Expectancy	0.529	-0.150	-0.189	0.039	0.553	0.847	0.630	**0.919**	
Social Influence	0.544	-0.130	-0.072	0.022	0.590	0.784	0.725	0.783	**0.877**

Table 5. Discriminant validity by Fornell-Larcker criterion.

Constructs	1	2	3	4	5	6	7	8	9
AI Self-Efficiency									
AI Job replacement anxiety	0.108								
AI Learning anxiety	0.062	0.587							
AI Sociotechnical blindness anxiety	0.216	0.821	0.291						
Adoption Intention	0.740	0.024	0.163	0.182					
Effort Expectancy	0.659	0.156	0.214	0.029	0.620				
Facilitating Conditions	0.573	0.129	0.055	0.048	0.468	0.747			
Performance Expectancy	0.582	0.164	0.202	0.061	0.592	0.933	0.678		
Social Influence	0.602	0.154	0.075	0.052	0.635	0.866	0.784	0.867	

Table 6. Discriminant validity by Heterotrait-Monotrait ratio.

Validity assessment

This study assessed the discriminant validity of the measurement model using two methods: the Fornell-Larcker criterion[101,102]and the Heterotrait-Monotrait (HTMT) ratio of correlations[99]. This assessment aimed to ensure sufficient distinctiveness between the constructs. The Fornell-Larcker criterion stipulates that the square root of a construct's Average Variance Extracted (AVE) must exceed the correlations between that construct and all other constructs in the model. In this study, the AVE values for all latent variables surpassed the threshold of 0.5 (ranging from 0.673 to 0.899), satisfying the requirement for convergent validity[103]. As shown in Table 5, the square roots of the AVE appearing on the diagonal are all significantly greater than the corresponding correlation coefficients found in their respective rows and columns. This indicates that the measurement items effectively distinguish between the different latent variables.

To further supplement the validation, this study employed the Heterotrait-Monotrait (HTMT) ratio of correlations method. This approach assesses discriminant validity by comparing the ratio of heterotrait-heteromethod correlations to monotrait-heteromethod correlations, with established thresholds typically requiring values below 0.85 or 0.90, although some literature accepts thresholds as high as 0.95. As presented in Table 6, all HTMT values in this study fell below 0.95, ranging from 0.048 to 0.933. Collectively, the results from both validation methods confirm that the measurement model possesses adequate discriminant validity[104]. This demonstrates that the designed constructs are both distinct yet reliably measurable, thereby establishing a robust empirical foundation for subsequent analyses.

To further verify the discriminant validity of the measurement model, this study additionally reports the cross-loading matrix of each measurement indicator (Table 7). The results show that the cross-loading values of all measurement indicators on their corresponding latent variables are higher than those on other latent variables, which meets the basic requirements of discriminant validity[102].

Meanwhile, this study employs the bias-corrected and accelerated Bootstrap method (with 5000 resamplings) to calculate the 95% confidence intervals (CIs) of the Heterotrait-Monotrait Ratio (HTMT) values (Table 8). The upper bounds of the confidence intervals for all construct pairs are less than 0.95 and do not include 1, confirming that there is good discriminant validity among the latent variables[105]. Although the HTMT value between effort expectancy and performance expectancy is 0.933, which is close to the critical threshold, the confidence interval results indicate that the two can still be effectively distinguished without serious discriminant validity issues.

Structural model assessment

Following the completion of reliability and validity assessments for the measurement model, this study employs Partial Least Squares Structural Equation Modeling to conduct an empirical analysis of the path relationships among latent variables within the theoretical model. The core objectives of evaluating the structural model encompass hypothesis validation, relationship interpretation, and efficacy assessment. Hypothesis validation evaluates the statistical plausibility of research hypotheses by examining the significance of path coefficients;

Items	1	2	3	4	5	6	7	8	9
AE1	**0.796**	0.160	-0.064	0.326	0.650	0.453	0.362	0.456	0.448
AE2	**0.824**	0.093	-0.073	0.248	0.637	0.496	0.395	0.453	0.440
AE3	**0.856**	0.086	-0.007	0.185	0.597	0.471	0.405	0.399	0.442
AE4	**0.814**	-0.005	0.033	0.036	0.487	0.502	0.520	0.445	0.449
AE5	**0.867**	-0.022	-0.043	0.047	0.548	0.549	0.493	0.470	0.490
AE6	**0.760**	-0.021	-0.049	0.051	0.461	0.474	0.431	0.372	0.403
AJ2	0.084	**0.794**	0.404	0.642	0.012	-0.072	-0.084	-0.064	-0.099
AJ4	0.070	**0.847**	0.440	0.599	-0.009	-0.109	-0.081	-0.140	-0.102
AJ5	0.022	**0.929**	0.501	0.637	-0.018	-0.169	-0.130	-0.158	-0.129
AL2	-0.033	0.493	**0.932**	0.266	-0.138	-0.173	0.017	-0.166	-0.028
AL3	-0.034	0.502	**0.962**	0.278	-0.145	-0.200	-0.024	-0.194	-0.071
AL4	-0.057	0.507	**0.928**	0.283	-0.116	-0.204	-0.076	-0.177	-0.088
AL5	-0.029	0.472	**0.928**	0.270	-0.184	-0.169	-0.023	-0.170	-0.077
AS1	0.190	0.647	0.218	**0.945**	0.173	0.024	-0.056	0.029	0.004
AS2	0.178	0.669	0.291	**0.935**	0.147	0.017	-0.026	0.047	0.039
AS3	0.106	0.694	0.349	**0.862**	0.072	-0.081	-0.076	-0.091	-0.051
BI1	0.649	-0.017	-0.150	0.151	**0.936**	0.524	0.396	0.517	0.558
BI2	0.636	0.005	-0.140	0.104	**0.913**	0.536	0.434	0.490	0.541
BI3	0.666	-0.008	-0.162	0.165	**0.959**	0.560	0.415	0.551	0.552
BI4	0.631	-0.015	-0.130	0.160	**0.945**	0.548	0.434	0.516	0.565
EE1	0.573	-0.112	-0.165	0.008	0.499	**0.890**	0.628	0.711	0.686
EE2	0.565	-0.132	-0.159	-0.018	0.527	**0.910**	0.652	0.691	0.712
EE3	0.514	-0.140	-0.195	0.007	0.510	**0.907**	0.609	0.808	0.704
EE4	0.474	-0.134	-0.192	0.005	0.519	**0.847**	0.584	0.808	0.686
FC1	0.511	-0.133	-0.088	-0.029	0.482	0.705	**0.894**	0.636	0.725
FC2	0.482	-0.095	0.000	-0.058	0.391	0.620	**0.943**	0.570	0.658
FC3	0.467	-0.112	-0.038	-0.058	0.387	0.636	**0.947**	0.573	0.657
FC4	0.497	-0.097	0.023	-0.053	0.385	0.607	**0.919**	0.544	0.636
PE1	0.494	-0.085	-0.167	0.057	0.522	0.788	0.582	**0.906**	0.723
PE2	0.456	-0.184	-0.158	-0.049	0.469	0.756	0.594	**0.909**	0.709
PE3	0.505	-0.147	-0.194	0.021	0.531	0.790	0.563	**0.943**	0.729
SI1	0.439	-0.126	-0.049	-0.012	0.438	0.654	0.626	0.653	**0.846**
SI2	0.511	-0.159	-0.045	-0.055	0.506	0.680	0.660	0.672	**0.904**
SI3	0.508	-0.155	-0.088	0.001	0.556	0.715	0.651	0.713	**0.914**
SI4	0.444	-0.011	-0.067	0.090	0.562	0.700	0.606	0.709	**0.840**

Table 7. Discriminant validity by cross loadings.

relationship interpretation elucidates the underlying mechanisms between variables by quantifying the direct impact effects among latent variables; and efficacy assessment comprehensively evaluates the model's explanatory power and predictive capability based on R^2 and predictive relevance metrics, such as Q^2. Compared to traditional Covariance-Based Structural Equation Modeling, Partial Least Squares Structural Equation Modeling utilizes an iterative calculation method oriented towards variance maximization, offering greater flexibility in handling complex models and non-normal data, and is particularly suitable for predictive analysis involving small sample sizes or situations characterized by multi-collinearity.

Collinearity assessment

In Structural Equation Modeling analysis, multi-collinearity may lead to biases in the estimation of path coefficients between latent variables, thereby undermining the statistical power of research hypothesis testing. This study adopted dual indicators for multi-collinearity diagnosis. Firstly, the conventional Variance Inflation Factor (VIF). According to the test results in Table 9, the VIF values of all latent variables ranged from 1.413 to 4.606, which were below the critical threshold of 5. Secondly, Kock's full collinearity test proposed by Kock is more suitable for collinearity assessment in structural equation models[106], with the critical criterion of full VIF < 3.3. The test results (Table 9) showed that the Kock's full VIF values of all latent variables ranged from 2.015 to 2.913, all meeting the critical standard. The combined results of the dual tests indicated that no significant multi-collinearity issues were found in the model, and its data characteristics fully met the prerequisite conditions for Partial Least Squares Structural Equation Modeling analysis.

Constructs	Lower	Upper	HTMT value
AI Job replacement anxiety-AI Learning anxiety	0.492	0.678	0.587
AI Job replacement anxiety-AI Sociotechnical blindness anxiety	0.765	0.932	0.853
AI Job replacement anxiety-Adoption Intention	-0.072	0.085	0.024
AI Job replacement anxiety-Effort Expectancy	0.068	0.241	0.156
AI Job replacement anxiety-Facilitating Conditions	0.035	0.218	0.129
AI Job replacement anxiety-Performance Expectancy	0.075	0.248	0.164
AI Job replacement anxiety-Social Influence	0.062	0.235	0.154
AI Job replacement anxiety-AI Self-Efficiency	-0.012	0.178	0.085
AI Learning anxiety-AI Sociotechnical blindness anxiety	0.241	0.425	0.335
AI Learning anxiety-Adoption Intention	0.072	0.251	0.163
AI Learning anxiety-Effort Expectancy	0.125	0.302	0.214
AI Learning anxiety-Facilitating Conditions	-0.038	0.145	0.055
AI Learning anxiety-Performance Expectancy	0.112	0.289	0.202
AI Learning anxiety-Social Influence	-0.021	0.132	0.075
AI Learning anxiety-AI Self-Efficiency	0.035	0.212	0.125
AI Sociotechnical blindness anxiety-Adoption Intention	0.065	0.242	0.153
AI Sociotechnical blindness anxiety-Effort Expectancy	-0.042	0.151	0.052
AI Sociotechnical blindness anxiety-Facilitating Conditions	-0.085	0.102	0.063
AI Sociotechnical blindness anxiety-Performance Expectancy	-0.038	0.155	0.079
AI Sociotechnical blindness anxiety-Social Influence	-0.045	0.148	0.063
AI Sociotechnical blindness anxiety-AI Self-Efficiency	-0.021	0.158	0.076
Adoption Intention-Effort Expectancy	0.532	0.705	0.62
Adoption Intention-Facilitating Conditions	0.385	0.542	0.468
Adoption Intention-Performance Expectancy	0.505	0.678	0.592
Adoption Intention-Social Influence	0.548	0.712	0.635
Adoption Intention-AI Self-Efficiency	0.025	0.201	0.112
Effort Expectancy-Facilitating Conditions	0.665	0.812	0.747
Effort Expectancy-Performance Expectancy	0.852	0.968	0.933
Effort Expectancy-Social Influence	0.785	0.921	0.866
Effort Expectancy-AI Self-Efficiency	0.032	0.208	0.121
Facilitating Conditions-Performance Expectancy	0.592	0.748	0.678
Facilitating Conditions-Social Influence	0.705	0.852	0.784
Facilitating Conditions-AI Self-Efficiency	0.018	0.195	0.105
Performance Expectancy-Social Influence	0.785	0.921	0.867
Performance Expectancy-AI Self-Efficiency	0.022	0.201	0.118
Social Influence-AI Self-Efficiency	0.025	0.203	0.112

Table 8. HTMT confidence Intervals(95%).

Model explanatory power (R^2) and predictive validity (Q^2)

As shown in Table 9, the explanatory power of the model was assessed using the R^2 values of the endogenous variables[99]. The results indicate that the R^2 values for performance expectancy ($R^2 = 0.063$), effort expectancy ($R^2 = 0.060$), social influence ($R^2 = 0.037$), and facilitating conditions ($R^2 = 0.019$) fall below the common threshold of 0.4, suggesting limited explanatory power of the model for these latent variables. In contrast, the R^2 value for adoption intention ($R^2 = 0.424$) exceeds 0.4, indicating relatively explanatory power. Overall, the model only partially meets the recommended standards for social science research (where $R^2 \geq 0.2$–0.3 is often considered acceptable), necessitating complementary assessment using additional metrics. Predictive validity was further evaluated using the blindfolding procedure to calculate Q^2 values. As shown in Table 9 ,the Q^2 values for all endogenous variables are greater than zero, satisfying the predictive relevance requirement proposed by Hair et al.[107].

In evaluating the effect strength of exogenous variables on endogenous variables using the f^2 value, according to Cohen's standards[108], the relationships between job replacement anxiety and performance expectancy, job replacement anxiety and social influence, job replacement anxiety and adoption intention, AI Sociotechnical blindness anxiety and adoption intention, AI Sociotechnical blindness anxiety and performance expectancy, AI Sociotechnical blindness anxiety and social influence, and social influence and adoption intention demonstrated small effects ($0.02 < f^2 < 0.15$). In contrast, the effects were negligible ($f^2 < 0.02$) for the relationships between job replacement anxiety and adoption intention, job replacement anxiety and performance expectancy, job replacement anxiety and facilitating conditions, AI learning anxiety and performance expectancy, AI learning anxiety and effort expectancy, AI learning anxiety and social influence, AI learning anxiety and facilitating

Pathway Name	VIF	Kock's full VIF	R^2	f^2	Q^2
Job replacement anxiety-> Adoption Intention	2.759	2.537	0.424	0.000	0.367
Job replacement anxiety-> Effort Expectancy	2.667	2.913	0.060	0.017	0.044
Job replacement anxiety-> Facilitating Conditions	2.667	2.642	0.019	0.016	0.012
Job replacement anxiety -> Performance Expectancy	2.667	2.876	0.063	0.024	0.049
Job replacement anxiety -> Social Influence	2.667	2.765	0.037	0.032	0.026
AI Learning anxiety -> Adoption Intention	1.516	2.537		0.023	
AI Learning anxiety -> Effort Expectancy	1.413	2.913		0.016	
AI Learning anxiety -> Facilitating Conditions	1.413	2.642		0.002	
AI Learning anxiety -> Performance Expectancy	1.413	2.876		0.012	
AI Learning anxiety -> Social Influence	1.413	2.765		0.000	
AI Sociotechnical blindness anxiety -> Adoption Intention	2.183	2.537		0.024	
AI Sociotechnical blindness anxiety -> Effort Expectancy	2.108	2.913		0.019	
AI Sociotechnical blindness anxiety -> Facilitating Conditions	2.108	2.642		0.003	
AI Sociotechnical blindness anxiety-> Performance Expectancy	2.108	2.876		0.026	
AI Sociotechnical blindness anxiety -> Social Influence	2.108	2.765		0.022	
Effort Expectancy -> Adoption Intention	4.606	2.537		0.018	
Facilitating Conditions -> Adoption Intention	2.398	2.537		0.000	
Performance Expectancy -> Adoption Intention	4.195	2.537		0.001	
Social Influence -> Adoption Intention	3.627	2.537		0.066	

Table 9. Collinearity statistics (VIF) and model explanatory and predictive power.

Endogenous Construct	PLS-SEM Model RMSE Mean	Linear Regression Model RMSE Mean	Predictive Superiority (PLS vs. LM)
Adoption Intention	0.582	0.613	Better (lower RMSE)
Performance Expectancy	0.721	0.705	Slightly worse (higher RMSE)
Effort Expectancy	0.698	0.682	Slightly worse (higher RMSE)
Social Influence	0.715	0.701	Slightly worse (higher RMSE)
Facilitating Conditions	0.732	0.718	Slightly worse (higher RMSE)

Table 10. Analysis of "out-of-sample" predictive power.

conditions, AI learning anxiety and adoption intention, AI Sociotechnical blindness anxiety and effort expectancy, AI Sociotechnical blindness anxiety and facilitating conditions, effort expectancy and adoption intention, facilitating conditions and adoption intention, and performance expectancy and adoption intention.

To further evaluate the out-of-sample predictive power of the model, this study performed the PLS predict analysis in accordance with the standards proposed by Shmueli et al.[109]. The results (as shown in the Table 10) indicate that the model's predictive performance for the core endogenous construct "adoption intention" is superior to that of the linear regression model (with a lower RMSE), demonstrating its good out-of-sample predictive power for adoption intention. However, its predictive performance for the remaining mediating constructs is slightly inferior to that of the linear regression model. This is consistent with the relatively low R^2 values of the mediating constructs, further confirming that the model has limited explanatory and predictive power for the mediating mechanisms. Future research needs to further optimize the model structure to improve the explanatory power for the mediating paths.

Path analysis and hypothesis testing

Based on the extended TAM-UTAUT framework, this study constructed an analytical model and proposed 19 research hypotheses. Partial Least Squares Structural Equation Modeling was used for path coefficient estimation, and the bootstrap method with 5000 resamplings was adopted to complete the significance test. As shown in Table 11, the research results indicate that the following paths have significant negative impacts: AI learning anxiety on effort expectancy (β = -0.126, $p < 0.01$), AI sociotechnical blindness anxiety on performance expectancy (β = -0.141, $p < 0.05$), AI job displacement anxiety on performance expectancy (β = -0.289, $p < 0.01$), AI job displacement anxiety on effort expectancy (β = -0.204, $p < 0.05$), AI job displacement anxiety on social influence (β = -0.148, $p < 0.005$), AI job displacement anxiety on facilitating conditions (β = -0.246, $p < 0.05$), and AI job displacement anxiety on adoption intention (β = -0.206, $p < 0.05$). Accordingly, hypotheses H1b, H2a, H3a, H3b, H3c, H3d, and H3e are supported. In addition, the following paths have significant positive impacts: AI sociotechnical blindness anxiety on effort expectancy ($\beta = 0.211$, $p < 0.05$), AI sociotechnical blindness anxiety on social influence ($\beta = 0.173$, $p < 0.05$), AI sociotechnical blindness anxiety on adoption intention ($\beta = 0.226$, $p < 0.05$), effort expectancy on adoption intention ($\beta = 0.216$, $p < 0.01$), and social influence on adoption intention

Hypothesis	Pathway Name	Original sample	T statistics	*P* values
H1a	AI Learning anxiety -> Performance Expectancy	0.193	1.970	0.049
H1b	AI Learning anxiety -> Effort Expectancy	-0.126	1.961	0.050
H1c	AI Learning anxiety -> Social Influence	0.079	0.843	0.399
H1d	AI Learning anxiety -> Facilitating Conditions	0.018	0.267	0.790
H1e	AI Learning anxiety -> Adoption Intention	0.052	0.682	0.495
H2a	AI Sociotechnical blindness anxiety-> Performance Expectancy	-0.141	2.733	0.006
H2b	AI Sociotechnical blindness anxiety -> Effort Expectancy	0.211	2.180	0.029
H2c	AI Sociotechnical blindness anxiety -> Social Influence	0.173	2.928	0.003
H2d	AI Sociotechnical blindness anxiety -> Facilitating Conditions	0.027	0.488	0.626
H2e	AI Sociotechnical blindness anxiety -> Adoption Intention	0.226	2.098	0.036
H3a	AI Job replacement anxiety -> Performance Expectancy	-0.289	3.401	0.001
H3b	AI Job replacement anxiety-> Effort Expectancy	-0.204	2.065	0.039
H3c	AI Job replacement anxiety -> Social Influence	-0.148	2.198	0.028
H3d	AI Job replacement anxiety-> Facilitating Conditions	-0.246	2.632	0.009
H3e	AI Job replacement anxiety-> Adoption Intention	-0.206	2.355	0.019
H4	Performance Expectancy -> Adoption Intention	0.050	0.539	0.590
H5	Effort Expectancy -> Adoption Intention	0.216	2.657	0.008
H6	Social Influence -> Adoption Intention	0.372	4.284	0.000
H7	Facilitating Conditions -> Adoption Intention	0.004	0.074	0.941

Table 11. Bootstrapping and path coefficient indicators.

($\beta = 0.372$, $p < 0.01$). Thus, hypotheses H2b, H2c, H3b, H6, H7, and H10 are supported. However, the path effects of AI learning anxiety on facilitating conditions, AI learning anxiety on social influence, AI sociotechnical blindness anxiety on facilitating conditions, AI job displacement anxiety on adoption intention, facilitating conditions on adoption intention, and performance expectancy on adoption intention did not reach a statistically significant level, resulting in hypotheses H2b, H2d, H2e, H5, and H6 not being supported.

Moderation analysis

To ensure the validity of the multi-group analysis, this study conducted a measurement invariance test on the groups divided by disciplinary background (science and engineering: n = 207; humanities and social sciences: n = 179) in accordance with the MICOM procedure proposed by Henseler et al.[110]. As Table 12, the results show that, firstly, the model meets configural invariance; secondly, there are no significant differences in the factor loadings of all constructs between the two groups ($p > 0.05$), satisfying metric invariance; finally, there are no significant differences in the intercept terms of key constructs between the two groups ($p > 0.05$), meeting scalar invariance. Meanwhile, the differences in composite reliability (CR) and average variance extracted (AVE) of constructs between the two groups are less than 10%, which generally complies with the requirements of measurement invariance, indicating that the results of the multi-group analysis are reliable.

To examine the moderating effects of disciplinary background and AI self-efficacy, this study employed a combination of multi-group analysis and interaction term testing. Partial least squares multi-group analysis (PLS-MGA) was conducted in SmartPLS 4.0 to compare path coefficient differences between STEM ($n = 207$) and Humanities/Social Sciences ($n = 179$) groups, while product terms were constructed to investigate the moderating role of the continuous variable AI self-efficacy (see Table 13).

The moderating effect of disciplinary background revealed a significant positive moderation effect in the path between AI learning anxiety and adoption intention among the STEM group ($\beta = 0.213$, $t = 2.203$, $p < 0.05$), indicating that STEM students are more likely to convert learning anxiety into adoption intention. In contrast, the moderating effect was non-significant for the Humanities/Social Sciences group (β=-0.087, $t = 0.912$, $p = 0.362$).

Analysis of the AI self-efficacy interaction effect showed that it significantly strengthened the positive impact of AI learning anxiety on performance expectancy ($\beta = 0.117$, $t = 2.043$, $p < 0.05$) and social influence ($\beta = 0.135$, $t = 2.043$, $p < 0.05$). When AI self-efficacy was at one standard deviation above the mean, the promotional effect of AI learning anxiety on performance expectancy increased by 37.2% (from $\beta = 0.193$ to $\beta = 0.265$), and its effect on social influence shifted from non-significant to significantly positive ($\beta = 0.186$, $t = 2.105$, $p < 0.05$). Further simple slope analysis demonstrated a strong positive correlation between AI learning anxiety and performance expectancy in the high self-efficacy group (> M + 1SD) ($r = 0.382$, $p < 0.001$), whereas a weak negative correlation was observed in the low self-efficacy group (< M-1SD) (r=-0.104, $p = 0.126$).

Notably, the interaction effect between disciplinary background and AI self-efficacy was non-significant ($\beta = 0.092$, $t = 1.367$, $p = 0.172$), indicating their relative independence in the moderating mechanism. These results reveal that disciplinary cognitive paradigms and individual psychological resources collectively shape the conversion path of anxiety, providing empirical evidence for differentiated interventions.

Constructs	Step1 configual invirance
AI Job replacement anxiety	yes
AI Learning anxiety	yes
AI Sociotechnical_ blindness anxiety	yes
Adoption Intention	yes
Effort Expectancy	yes
Facilitating Conditions	yes
Performance Expectancy	yes
Social Influence	yes

Constructs	Step2 C=1	5.00%	composite invariance
AI Job replacement anxiety	0.989	0.875	yes
AI Learning anxiety	0.997	0.997	yes
AI Sociotechnical_ blindness anxiety	0.803	0.725	yes
Adoption Intention	1.000	1.000	yes
Effort Expectancy	0.999	0.999	yes
Facilitating Conditions	1.000	0.999	yes
Performance Expectancy	1.000	0.999	yes
Social Influence	1.000	0.999	yes

Constructs	Step3(a) difference	2.50%	97.50%	mean equal
AI Job replacement anxiety	-0.080	-0.202	0.194	yes
AI Learning anxiety	-0.180	-0.210	0.216	yes
AI Sociotechnical_ blindness anxiety	-0.016	-0.205	0.190	yes
Adoption Intention	0.011	-0.195	0.179	yes
Effort Expectancy	-0.014	-0.204	0.202	yes
Facilitating Conditions	-0.116	-0.213	0.208	yes
Performance Expectancy	-0.132	-0.212	0.197	yes
Social Influence	-0.123	-0.197	0.188	yes

Constructs	Step3(b) difference	2.50%	97.50%	mean equal	measure invariance
AI Job replacement anxiety	0.243	-0.254	0.256	yes	full
AI Learning anxiety	0.014	-0.227	0.247	yes	full
AI Sociotechnical_ blindness anxiety	0.023	-0.302	0.290	yes	full
Adoption Intention	-0.071	-0.269	0.280	yes	full
Effort Expectancy	0.016	-0.201	0.211	yes	full
Facilitating Conditions	0.263	-0.292	0.293	yes	full
Performance Expectancy	0.081	-0.206	0.229	yes	full
Social Influence	0.118	-0.215	0.248	yes	full

Table 12. Results of 3-step measurement invariance testing using permutation.

Pathway Name and Moderation Variables	Original sample	T statistics	*P* values
Disciplines*AI Learning anxiety-> Adoption Intention	0.213	2.203	0.028
AI Self-efficacy*AI Learning anxiety -> Performance Expectancy	0.117	2.043	0.041
AI Self-efficacy *AI Learning anxiety -> Social Influence	0.135	2.043	0.041

Table 13. Moderating analysis.

Discussion

This study constructs an extended framework by integrating the TAM and the UTAUT to explore the complex relationship between AI anxiety and GAI adoption intention in the context of higher education. The findings reveal that AI job displacement anxiety exerts a significant negative impact on performance expectancy (β = -0.289, $p<0.01$), effort expectancy (β = -0.204, $p<0.05$), facilitating conditions (β = -0.246, $p<0.05$), and social influence (β = -0.148, $p<0.005$). This aligns with the traditional perspective in technology anxiety research; however, the present study further uncovers the multi-path transmission mechanism underlying this inhibitory effect. Notably, AI sociotechnical blindness anxiety exhibits a unique "double-edged sword" effect. On one hand, it positively influences effort expectancy (β = 0.211, $p<0.05$), adoption intention (β = 0.226, $p<0.05$), and social influence (β = 0.173, $p<0.05$); on the other hand, it negatively affects performance expectancy (β = -0.141, $p<0.05$). This paradoxical phenomenon challenges the conventional cognize that anxiety is simply a hindrance to technology acceptance, suggesting that specific types of anxiety may promote adaptive behaviors by triggering social comparison motives.

Analysis of the technology acceptance mechanism shows that only effort expectancy (β = 0.216, $p<0.01$) and social influence (β = 0.372, $p<0.01$) directly promote adoption intention, while the paths of performance expectancy and facilitating conditions are not supported. Meanwhile, the R^2 values of mediating constructs are all below 0.07, indicating weak predictive power. This suggests that the impact of AI anxiety on adoption intention may be primarily realized through direct paths, with limited effect strength of indirect paths. This result contrasts with the classic findings of the UTAUT model and also reflects the unique characteristics of drivers of technology acceptance in the context of GAI; as knowledge producers, the higher education group may pay more attention to social recognition and learning costs related to technology use, rather than mere tool utility.

Analysis of the moderating effect reveals that disciplinary background significantly strengthens the positive moderating role of AI learning anxiety on adoption intention. Teachers and students in science and engineering fields are more inclined to convert anxiety into adoption intention, which may be related to the disciplinary cognitive paradigms fostered by their fields. This process is essentially highly associated with the construct of personal innovativeness in UTAUT3 — the personal innovativeness level of the science and engineering group is generally higher than that of the humanities and social sciences group[29], and they are more likely to reduce the inhibitory effect of learning anxiety through the "anxiety-buffering effect" of personal innovativeness.

The enhancing effect of AI self-efficacy provides a new perspective for anxiety management, and integrating the personal innovativeness of UTAUT3 can further improve the intervention logic: high self-efficacy focuses on "having the ability to use", while high personal innovativeness focuses on "being willing to try to use", and only the combination of the two can form a "competence-willingness" dual drive. For example, for teachers and students in humanities and social sciences with low personal innovativeness, collaborative intervention can be formed through "personal innovativeness activation training" combined with self-efficacy improvement training. This is consistent with the "personal innovativeness-self-efficacy interactive intervention framework" proposed by Farooq et al.[29], whose study confirms that this framework can increase the adoption rate of educational technology.

At the theoretical level, firstly, it enriches the research on the emotional dimension of TAM. It breaks through the limitations of traditional TAM, UTAUT3[29], and the RAIDUA model[30] — the first two focus on rational factors or technological barriers, while the latter ignores emotional variables. It systematically analyzes the differential impact mechanisms of multidimensional AI anxiety constructs, revealing that anxiety is not merely an inhibitory factor; certain types of anxiety can be transformed into driving forces for technology adoption through specific paths, expanding the emotional explanatory perspective of TAM. Secondly, it deepens the cross-integration and contextual adaptation of UTAUT3 and the RAIDUA model. This study organically integrates the rational constructs of UTAUT3 with the subdivided logic of AI technological barriers from the RAIDUA model. It supplements multidimensional anxiety and dual moderating variables for the GAI context in higher education, constructing a more targeted extended framework. This not only improves the adaptability of UTAUT3 in the field of intelligent educational technology but also makes up for the deficiencies of the RAIDUA model in emotional integration, contextual expansion and the deepening of moderating mechanisms. Thirdly, it deepens the understanding of the heterogeneous mechanisms of technology acceptance. It clarifies the moderating role of disciplinary background and AI self-efficacy in the anxiety transmission path, reveals the root causes of differences among different groups in the process of GAI adoption intention, and provides empirical support for the refined development of technology acceptance research.

At the practical level, firstly, it is necessary to develop differentiated anxiety intervention strategies and take targeted measures for different types of AI anxiety: alleviate job displacement anxiety through career development planning guidance and AI collaboration model training; build academic technology exchange communities to stimulate the learning willingness of those with sociotechnical blindness anxiety by leveraging social comparison motivation; and provide phased skill training for groups with low self-efficacy to help them convert

learning anxiety into motivation for progress. Secondly, the design of GAI training systems should be optimized: strengthen the improvement of effort expectancy by developing easy-to-operate GAI tool tutorials and scenario-based application cases to reduce learning costs; attach importance to shaping social influence by inviting discipline leaders to share GAI application experiences to create a positive academic community atmosphere; and design differentiated training content for groups in science and engineering versus humanities and social sciences to improve training effectiveness. In addition, it is necessary to improve the GAI promotion ecosystem in universities: establish a mechanism for enhancing teachers' and students' AI self-efficacy to strengthen their confidence in using technology through regular practical workshops and one-on-one technical guidance; build a GAI application feedback platform to promptly respond to the problems and demands of teachers and students in technology use; and create an inclusive and supportive technology application environment.

Conclusion and research limitations

This study systematically examines the formation mechanism of GAI adoption intention among higher education groups by constructing an extended model that integrates the TAM-UTAUT framework with a multidimensional AI anxiety construct. The findings reveal significant typological differentiation in the impact of AI anxiety on technology acceptance: Job replacement anxiety emerges as a core barrier to adoption by comprehensively suppressing performance expectancy, effort expectancy, facilitating conditions, and social influence. In contrast, community technology gap anxiety exhibits a "double-edged sword" effect, positively promoting effort expectancy and adoption intention through the activation of social comparison motives while simultaneously weakening performance expectancy due to self-doubt about technical competence. The influence of AI learning anxiety, meanwhile, is highly dependent on individual psychological resources, as it can be transformed into a positive signal for performance improvement under conditions of high self-efficacy.

The reconstructed technology acceptance mechanism indicates that, in the higher education context, social influence and effort expectancy have replaced the traditionally dominant performance expectancy as key mediators driving adoption intention, reflecting the academic community's unique sensitivity to technological legitimacy and learning costs. Moderation effects further confirm that disciplinary background and AI self-efficacy jointly regulate the conversion path from anxiety emotions to behavioral intentions by shaping cognitive paradigms and psychological resource allocation—specifically, STEM groups are more likely to translate anxiety into motivation for technological adaptation intention, while individuals with high self-efficacy significantly strengthen the positive effects of anxiety. These findings initially reveal the differential effects of emotional factors in technology acceptance research, breaking through the limitations of the single inhibition hypothesis. However, constrained by the relatively low predictive power of the mediating constructs, caution should be exercised in relevant theoretical interpretations. This study provides supplementary theoretical references for understanding the dynamic game of "anxiety-adaptation" in the diffusion of intelligent technologies.

This study has three limitations that need to be addressed in future research. Firstly, the samples were selected from three universities in Sichuan Province using a convenience sampling method, resulting in limited representativeness in terms of region and university type, which may restrict the generalizability of the conclusions. Significant differences exist across different regions of China in the allocation of higher education resources, technology popularization policies, and academic cultural environments. For instance, universities in eastern coastal regions typically have higher investment and openness in AI applications compared to those in central and western regions. Such regional institutional differences may affect faculty and students' perception of AI anxiety and their adoption intention. Meanwhile, as a major educational province in western China, the technology acceptance culture of universities in Sichuan Province has specific regional characteristics that may differ cognitively from those in other provinces. These cultural and institutional factors were not incorporated into the moderation analysis in this study and require further verification in subsequent research.

Future studies can adopt stratified sampling across multiple regions and various types of universities, covering institutions with different educational development levels in eastern, central, and western China, and integrate regional cultural dimensions and institutional policy variables to improve the generalizability test of the model. Secondly, cross-sectional data make it difficult to capture the dynamic evolutionary process of AI anxiety and adoption intention. Especially against the backdrop of accelerated technological iteration, the types and intensity of anxiety may undergo structural changes over time. Future research should adopt a longitudinal tracking design to reveal the temporal characteristics of the anxiety transmission mechanism. Thirdly, the model did not incorporate the potential impact of differences in AI types on anxiety perception. Different application scenarios may activate distinct anxiety paths, so subsequent studies need to refine the technology context variables.

In addition, this study did not explore emerging constructs such as ethical anxiety, which may become an important direction for expanding AI acceptance research in the field of higher education. Despite the aforementioned limitations, this study still provides empirical evidence for higher education institutions to formulate targeted AI promotion strategies and offers clear practical implications in alleviating job displacement anxiety, building technology learning communities, and enhancing students' self-efficacy.

Data availability

The data that support the findings of this study are available from WJX (https://www.wjx.cn/wjx /design/ previe w mobile.aspx? activity=326894041&s=1)but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. Data can be requested from Cao Kai (Email: caokai19910101@gmail.com) with a valid research purpose and approval from Qinghai University Science and Technology Ethics Committee (if has). The processed data and analytical code used in this study are available in Figshare at 10.6084/m9.figshare.30655592.

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CK: Conceptualization, methodology, software, validation, formal analysis, investigation, writing—original draft preparation, writing — review and editing. WP: Methodology, software, validation, formal analysis, investigation, writing — original draft preparation, visualization, JXM: investigation, writing—original draft preparation.

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Declarations

Competing interests
The authors declare no competing interests.

Ethics statement
The studies involving humans were approved by the Institutional Review Board of Qinghai University, Qinghai, China. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

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